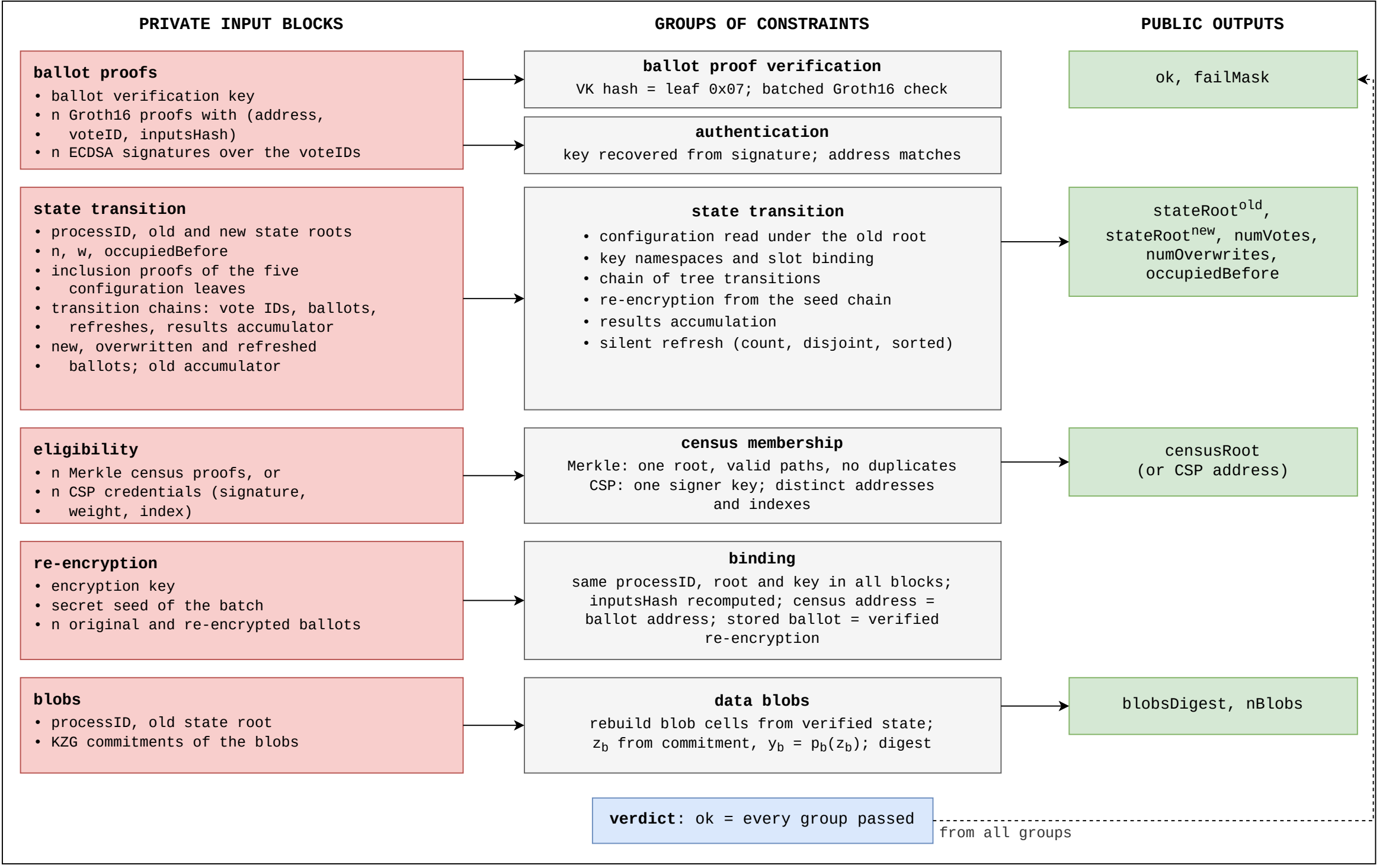
PRIVATE INPUT BLOCKS
GROUPS OF CONSTRAINTS
PUBLIC OUTPUTS
ballot proofs
• ballot verification key
• n Groth16 proofs with (address,
• voteID, inputsHash)
• n ECDSA signatures over the voteIDs
ballot proof verification
VK hash = leaf 0x07; batched Groth16 check
authentication
key recovered from signature; address matches
ok, failMask
state transition
• processID, old and new state roots
• n, w, occupiedBefore
• inclusion proofs of the five
• configuration leaves
• transition chains: vote IDs, ballots,
• refreshes, results accumulator
• new, overwritten and refreshed
• ballots; old accumulator
state transition
• configuration read under the old root
• key namespaces and slot binding
• chain of tree transitions
• re-encryption from the seed chain
• results accumulation
• silent refresh (count, disjoint, sorted)
stateRoot^old, stateRoot^new, numVotes, numOverwrites, occupiedBefore
eligibility
• n Merkle census proofs, or
• n CSP credentials (signature,
• weight, index)
census membership
Merkle: one root, valid paths, no duplicates
CSP: one signer key; distinct addresses and indexes
censusRoot (or CSP address)
re-encryption
• encryption key
• secret seed of the batch
• n original and re-encrypted ballots
binding
same processID, root and key in all blocks; inputsHash recomputed; census address = ballot address; stored ballot = verified re-encryption
blobs
• processID, old state root
• KZG commitments of the blobs
data blobs
rebuild blob cells from verified state; z_b from commitment, y_b = p_b(z_b); digest
blobsDigest, nBlobs
verdict: ok = every group passed
from all groups